

Cybersecurity Bootcamp – Frequently Asked Questions

Purpose

This document is a running collection of frequently asked questions, important concepts, commands, and lessons from class. New questions and answers can be added each day as topics are covered.

Table of Contents

1. General Cybersecurity
 2. Networking
 3. Linux
 4. Windows
 5. Access Control & Permissions
 6. System Hardening
 7. Vulnerabilities & Threats
 8. Governance, Risk & Compliance (GRC)
 9. Cloud & Virtualization
 10. Incident Response
 11. Business Continuity & Disaster Recovery
 12. Tools & Commands
 13. Labs & Troubleshooting
 14. Miscellaneous Questions
-

1. General Cybersecurity

Question: What is PKI, and how do digital certificates and signatures establish trust?

Answer: PKI (Public Key Infrastructure) is the system of certificates, public/private keys, certificate authorities, and policies used to establish trusted digital identities. A digital

certificate connects a public key to a verified identity, while a digital signature uses a private key to prove who signed the data and confirm that it has not been altered.

Category: General Cybersecurity

Key Takeaway: PKI creates a trusted system for managing digital identities, while certificates verify who owns a public key and digital signatures provide authenticity, integrity, and non-repudiation.

Question: What is the difference between symmetric, asymmetric, and hybrid encryption?

Answer: Symmetric encryption uses the same secret key to encrypt and decrypt data and is generally very fast. Asymmetric encryption uses a public and private key pair and is useful for secure key exchange and digital signatures. Hybrid encryption combines both methods by using asymmetric encryption to securely exchange a symmetric key, then using symmetric encryption for the actual data.

Category: General Cybersecurity

Key Takeaway: Symmetric encryption is fast, asymmetric encryption helps establish trust and exchange keys securely, and hybrid encryption combines the strengths of both.

Question: What is the difference between RBAC and ABAC?

Answer: RBAC (Role-Based Access Control) gives permissions based on a user's assigned role, such as administrator, analyst, or employee. ABAC (Attribute-Based Access Control) makes access decisions using attributes such as department, location, device type, clearance level, or time of day.

Category: Access Control & Permissions

Key Takeaway: RBAC uses roles, while ABAC uses attributes and conditions to make more detailed access decisions.

Question: What is the difference between a traditional firewall and a next-generation firewall (NGFW)?

Answer: A traditional firewall mainly makes decisions based on information such as IP addresses, ports, protocols, and connection state. A next-generation firewall (NGFW)

adds deeper inspection features such as application awareness, intrusion prevention, threat intelligence, identity-based policies, and SSL/TLS inspection.

Category: General Cybersecurity

Key Takeaway: Traditional firewalls focus mostly on where traffic is coming from, where it is going, and what port it uses, while NGFWs can also inspect what the traffic actually is and whether it behaves maliciously.

2. Networking

Question: What is the difference between a MAC address and an IP address?

Answer: A MAC address is a hardware address used to identify a network interface on the local network, while an IP address is a logical address used to identify a device and route traffic between networks. MAC addresses operate mainly at Layer 2, while IP addresses operate at Layer 3.

Category: Networking

Key Takeaway: MAC addresses identify devices locally, while IP addresses are used to route traffic across networks.

Question: What information does a computer need from DHCP to communicate properly on a network?

Answer: DHCP commonly provides the computer with an IP address, subnet mask, default gateway, and DNS server. These settings allow the device to identify itself, determine which devices are local, reach other networks, and resolve domain names.

Category: Networking

Key Takeaway: The four main DHCP settings to remember are IP address, subnet mask, default gateway, and DNS server.

Question: Why should traffic be monitored at every LAN interconnection point?

Answer: LAN interconnection points are places where traffic moves from one network segment to another, making them important locations for detecting suspicious activity.

Monitoring these points can reveal unauthorized access, unusual traffic patterns, malware communication, and attempts at lateral movement.

Category: Networking

Key Takeaway: Monitoring every LAN interconnection helps prevent visibility gaps and makes it easier to detect threats moving between networks.

Question: How does network segmentation reduce the risk of lateral movement?

Answer: Network segmentation divides a network into smaller zones or segments and controls the traffic allowed between them. If an attacker compromises one system, segmentation can prevent or limit them from freely moving into other parts of the network.

Category: Networking

Key Takeaway: Segmentation limits an attacker's ability to move from one compromised area to another, reducing the potential blast radius of an attack.

Question: Why is centralized management important in a multi-LAN environment?

Answer: Centralized management gives administrators one place to monitor multiple LANs, apply consistent security policies, and make configuration changes. It also helps prevent policy drift, where different networks slowly end up with different or outdated security settings.

Category: Networking

Key Takeaway: Centralized management improves visibility, consistency, and control across all connected LANs.

3. Linux

Question: What is a mount point?

Answer: A mount point is a directory in Linux where a storage device or file system is attached so its files can be accessed. For example, a USB drive might be mounted to a directory such as `/mnt/usb`.

Category: Linux

Key Takeaway: A mount point is the location in the Linux directory structure where a file system or storage device becomes accessible.

Question:

Answer:

Command(s):

Key Takeaway:

Notes:

4. Windows

Question: What mechanism is commonly used to push configuration changes to Windows servers?

Answer: Group Policy is commonly used to centrally deploy and enforce configuration changes across Windows computers and servers in an Active Directory environment. Administrators create or update Group Policy Objects (GPOs), which are then applied to selected users, computers, organizational units, or domains.

Category: Windows

Key Takeaway: Group Policy and GPOs let administrators centrally push and enforce Windows configuration changes across multiple systems.

Question:

Answer:

Command / Tool:

Key Takeaway:

Notes:

5. Access Control & Permissions

Question: What is an active directory?

Answer: Active Directory is Microsoft's directory service used to manage users, computers, groups, permissions, and other resources on a Windows network. It allows administrators to centrally control authentication and access across an organization.

Category: Access Control & Permissions

Key Takeaway: Active Directory gives organizations a central place to manage identities, devices, and access.

Question: What is the Kerberos authentication process?

Answer: Kerberos is an authentication protocol that uses encrypted tickets instead of repeatedly sending a user's password across the network. First, the user logs in and receives a Ticket Granting Ticket (TGT). The TGT is then used to request a service ticket, which is presented to the server to prove the user's identity and gain access.

Category: Access Control & Permissions

Key Takeaway: Kerberos follows the basic process Login → TGT → Service Ticket → Access and helps protect passwords by using tickets for authentication.

Question: What is a jump host?

Answer: A jump host, or jump server, is a secured system that administrators connect to first before accessing sensitive servers or internal systems. It acts as a controlled gateway and helps limit, monitor, and log administrative access.

Category: Access Control & Permissions

Key Takeaway: A jump host provides a secure, centralized entry point for accessing sensitive systems instead of allowing direct administrative connections.

Question: What are the most important MFA enforcement and recovery strategies?

Answer: MFA should be enforced for primary logins, privileged or administrative actions, and remote access. Common enforcement methods include conditional access policies, IAM-integrated requirements, and application-level enforcement. Organizations should also maintain secure recovery options such as backup codes, controlled recovery workflows, and carefully protected break-glass accounts.

Category: Access Control & Permissions

Key Takeaway: Strong MFA requires both consistent enforcement across important systems and secure recovery methods that do not create an easy way to bypass MFA.

Question: What is the difference between SSO, MFA, and a one-time password?

Answer: SSO (Single Sign-On) allows a user to authenticate once and access multiple approved applications. MFA (Multi-Factor Authentication) requires two or more different authentication factors, such as a password and a phone notification. A one-time password (OTP) is a temporary code that can only be used once and is often used as one factor in MFA.

Category: Access Control & Permissions

Key Takeaway: SSO reduces the number of logins, MFA increases authentication security, and an OTP is a temporary credential that can be used as part of MFA.

Question: What is the difference between RBAC and ABAC?

Answer: RBAC (Role-Based Access Control) gives permissions based on a user's assigned role, such as administrator, analyst, or employee. ABAC (Attribute-Based Access Control) makes access decisions using attributes such as department, location, device type, clearance level, or time of day.

Category: Access Control & Permissions

Key Takeaway: RBAC uses roles, while ABAC uses attributes and conditions to make more detailed access decisions.

6. System Hardening

Question: Why are firewall rules and configuration just as important as the firewall's technical capabilities?

Answer: Advanced firewall features only provide protection when the rules and policies are configured to use them correctly. Poorly written or overly permissive rules can allow unauthorized traffic, lateral movement, or data exfiltration even when the firewall itself has advanced security capabilities.

Category: System Hardening

Key Takeaway: A powerful firewall is only as effective as its configuration; bad rules can make advanced security features ineffective.

Question: Why does the order of firewall rules matter?

Answer: Firewall rules are usually processed from top to bottom, and the first matching rule is applied. Because of this, a broad rule placed above a more specific rule can cause the specific rule to never be reached.

Category: System Hardening

Key Takeaway: Firewall rule order matters because the first matching rule usually wins, so more specific rules should be placed carefully to avoid unintended access or blocking.

Question: What is the difference between default deny and default allow?

Answer: Default deny blocks traffic unless it is specifically permitted by a rule. Default allow permits traffic unless it is specifically blocked. Default deny is generally more secure because only approved traffic is allowed through.

Category: System Hardening

Key Takeaway: Default deny = block everything unless approved; default allow = allow everything unless blocked.

Question: What is a shadow rule, and why can it be dangerous?

Answer: A shadow rule is a firewall rule that never gets used because an earlier rule already matches the same traffic. This can be dangerous because administrators may think the shadowed rule is providing protection when it is actually being ignored.

Category: System Hardening

Key Takeaway: A shadow rule is hidden behind a broader earlier rule, which can create unexpected access, security gaps, and difficult troubleshooting.

Question: Why should firewall rule updates and network segmentation be used together?

Answer: Firewall rule updates help block known threats, while network segmentation limits how far an attacker can move if something gets through. Using both provides layered protection because one control helps cover the weaknesses of the other.

Category: System Hardening

Key Takeaway: Updated rules block known threats, while segmentation limits the damage from unknown or successful attacks.

7. Vulnerabilities & Threats

Question: Three most common password cracking tools?

Answer: John the Ripper, Hashcat, Hydra

Category: Vulnerabilities & Threats

Key Takeaway: John the Ripper and Hashcat are mainly used to crack password hashes, while Hydra is commonly used to test login credentials against network services.

Question: What is an NBNS/LLMNR poisoning attack?

Answer: An NBNS/LLMNR poisoning attack happens when an attacker responds to local network name-resolution requests before the legitimate system does. The victim may then connect to the attacker's machine and send authentication information, such as an NTLM challenge-response, which the attacker may try to capture or relay.

Category: Vulnerabilities & Threats

Key Takeaway: NBNS/LLMNR poisoning tricks devices into trusting a malicious system during local name resolution, which can expose or relay Windows authentication credentials.

Question: What is lateral movement?

Answer: Lateral movement is when an attacker who has already compromised one system moves through the network to access additional systems, accounts, or resources. Attackers often use stolen credentials, remote access tools, or trusted network connections to move between systems.

Category: Vulnerabilities & Threats

Key Takeaway: Lateral movement allows an attacker to expand from one compromised system to other parts of the network.

Question: What is ARP poisoning, and why is ARP vulnerable to it?

Answer: ARP poisoning is an attack where a malicious device sends false ARP information so other systems associate the attacker's MAC address with another device's IP address, such as the default gateway. ARP is vulnerable because it does not authenticate ARP replies and devices can accept unsolicited updates.

Category: Vulnerabilities & Threats

Key Takeaway: ARP poisoning works because ARP trusts mapping updates without verifying that they are legitimate.

Question: How does an NGFW detect malicious traffic that is hidden inside normal HTTPS traffic on port 443?

Answer: An NGFW can use SSL/TLS inspection to decrypt HTTPS traffic, inspect its contents and behavior, and then re-encrypt it before forwarding it. It can also use application identification, intrusion prevention, and behavioral analysis to detect traffic that looks like normal HTTPS but is actually malware or command-and-control communication.

Category: Vulnerabilities & Threats

Key Takeaway: An NGFW does not automatically trust traffic just because it uses HTTPS or port 443; it can decrypt and inspect the traffic to determine what it is actually doing.

Question: What are common signs of firewall evasion?

Answer: Common signs include traffic using unusual or non-standard ports, repeated blocked connection attempts, unexpected protocol behavior, suspicious encrypted traffic, port scanning, and applications trying to disguise themselves as normal allowed traffic.

Category: Vulnerabilities & Threats

Key Takeaway: Firewall evasion often appears as unusual ports, repeated denied traffic, protocol anomalies, or malicious traffic trying to blend in with legitimate connections.

8. Governance, Risk & Compliance (GRC)

Question:

Answer:

Framework / Standard:

Why It Matters:

Notes:

Question:

Answer:

Framework / Standard:

Why It Matters:

Notes:

9. Cloud & Virtualization

Question: What is IaaS? (Infrastructure as a Service)

Answer: Infrastructure as a Service (IaaS) is a cloud computing model where a provider supplies virtualized resources such as servers, storage, and networking. The customer typically manages the operating systems, applications, and configurations running on that infrastructure.

Category: Cloud & Virtualization

Key Takeaway: IaaS provides the underlying computing infrastructure while giving the customer control over the operating systems and applications.

Question: What is east-west VM communication?

Answer: East-west VM communication refers to network traffic moving between virtual machines or systems within the same internal environment, such as within a data center or cloud network.

Category: Cloud & Virtualization

Key Takeaway: East-west traffic stays inside the environment, while north-south traffic generally moves into or out of the environment.

Question: What is IaaS? (Infrastructure as a Service)

Answer: Infrastructure as a Service (IaaS) is a cloud computing model where a provider supplies virtualized resources such as servers, storage, and networking. The customer typically manages the operating systems, applications, and configurations running on that infrastructure.

Category: Cloud & Virtualization

Key Takeaway: IaaS provides the underlying computing infrastructure while giving the customer control over the operating systems and applications.

Question: Out of the Threat Mitigation Framework layers, which layer is most important for preventing VM Escape?

Answer: Hypervisor hardening is the most important layer for preventing VM Escape. VM Escape occurs when an attacker breaks out of a guest virtual machine and gains access to the hypervisor or host system, so securing and patching the hypervisor directly addresses that risk.

Category: Cloud & Virtualization

Key Takeaway: Because VM Escape targets the boundary between the VM and the hypervisor, hypervisor hardening is the primary defense.

Question: Why is Ring -1 important to protect?

Answer: Ring -1 refers to the hypervisor layer, which sits below the operating system and controls virtual machines and access to the underlying hardware. If an attacker compromises the hypervisor, they could potentially affect multiple VMs at once, bypass protections inside individual operating systems, and gain very powerful control over the virtual environment.

Category: Cloud & Virtualization

Key Takeaway: Ring -1 is critical to protect because compromising the hypervisor can give an attacker control over multiple virtual machines and the systems running on them.

Question: What hypervisor has a smaller attack surface, Type 1 or Type 2? Why?

Answer: A Type 1 hypervisor generally has a smaller attack surface because it runs directly on the physical hardware instead of on top of a host operating system. A Type 2 hypervisor depends on the security of both the hypervisor software and the underlying host OS, creating more components that could potentially be attacked.

Category: Cloud & Virtualization

Key Takeaway: Type 1 hypervisors usually have a smaller attack surface because they eliminate the extra host operating system layer.

Question: What is the difference between a virtual machine and a container?

Answer: A virtual machine includes its own full operating system and runs through a hypervisor, while a container shares the host system's operating system kernel and mainly packages the application and its dependencies. Containers are generally more lightweight and faster to start than VMs.

Category: Cloud & Virtualization

Key Takeaway: VMs virtualize entire operating systems, while containers virtualize applications while sharing the host OS kernel.

Question: What is the difference between Type 1 and Type 2 Hypervisors?

Answer: A Type 1 hypervisor runs directly on the physical hardware, while a Type 2 hypervisor runs on top of a normal operating system. Type 1 hypervisors are generally used in enterprise environments because they offer better performance, security, and resource management.

Category: Cloud & Virtualization

Key Takeaway: Type 1 runs directly on hardware, while Type 2 depends on a host operating system.

Question: What are 4 ways we can have shared storage?

Answer: Four common shared storage methods are SAN (Storage Area Network), NAS (Network Attached Storage), iSCSI, and Fibre Channel. These allow multiple servers or hosts to access centralized storage instead of relying only on local disks.

Category: Cloud & Virtualization

Key Takeaway: Shared storage lets multiple hosts access the same centralized data, which is important for clustering, migration, and high availability.

Question: What is the difference between Fault Tolerance (FT) and High Availability (HA)?

Answer: Fault Tolerance keeps a duplicate system running at the same time so service can continue with little or no interruption if the primary system fails. High Availability

detects a failure and restarts or moves the workload to another available host, which may cause a short period of downtime.

Category: Cloud & Virtualization

Key Takeaway: FT aims for near-zero downtime, while HA focuses on quick recovery after a failure.

Question: What is the difference between a bastion host and a jump host?

Answer: A jump host is a controlled system administrators use to access other internal systems. A bastion host is a specially hardened system designed to be exposed to or positioned near an untrusted network and act as a secure access point. In practice, a bastion host can also function as a jump host.

Category: Access Control & Permissions

Key Takeaway: A jump host focuses on controlling administrative access, while a bastion host emphasizes hardened security at the boundary of a trusted network.

Question: What is Infrastructure as Code (IaC)?

Answer: Infrastructure as Code is the practice of managing and configuring infrastructure through code or configuration files instead of making changes manually. It allows environments to be deployed consistently, repeatedly, and automatically.

Category: Cloud & Virtualization

Key Takeaway: IaC makes infrastructure easier to automate, standardize, and recreate consistently.

Question: How do you isolate a machine in a virtual environment?

Answer: A virtual machine can be isolated by placing it on a separate virtual network, VLAN, or subnet and restricting communication with firewall rules, access controls, or virtual switch settings. The goal is to limit what other systems the VM can communicate with.

Category: Cloud & Virtualization

Key Takeaway: VM isolation uses network segmentation and access controls to prevent unnecessary communication between systems.

Question: Why are virtual firewall appliances useful in cloud and virtualized environments?

Answer: Virtual firewall appliances provide firewall capabilities without requiring dedicated physical hardware. They can be deployed inside cloud or virtual environments to inspect traffic between virtual machines, networks, and applications, making them easier to scale and place where traffic actually flows.

Category: Cloud & Virtualization

Key Takeaway: Virtual firewalls bring security controls directly into virtual and cloud environments where physical firewalls may not see all traffic.

10. Incident Response

Question:

Answer:

Incident Response Stage:

Key Takeaway:

Notes:

Question:

Answer:

Incident Response Stage:

Key Takeaway:

Notes:

11. Business Continuity & Disaster Recovery

Question:

Answer:

Key Term / Concept:

Example:

Notes:

Question:

Answer:

Key Term / Concept:

Example:

Notes:

12. Tools & Commands

Command / Tool

Name:

Purpose:

Syntax / Command:

Example:

Important Notes:

Command / Tool

Name:

Purpose:

Syntax / Command:

Example:

Important Notes:

13. Labs & Troubleshooting

Lab / Issue

Date:

Lab:

Problem:

What Caused It:

Solution:

Command(s) Used:

What was Learned:

Lab / Issue

Date:

Lab:

Problem:

What Caused It:

Solution:

Command(s) Used:

What was Learned:

14. Miscellaneous Questions

Question:

Answer:

Key Takeaway:

Notes:

Daily Additions

Date: 8-13-2026

Question 1

Question: What is an active directory?

Answer: Active Directory is Microsoft's directory service used to manage users, computers, groups, permissions, and other resources on a Windows network. It allows administrators to centrally control authentication and access across an organization.

Category: Access Control & Permissions

Key Takeaway: Active Directory gives organizations a central place to manage identities, devices, and access.

Question 2

Question: Three most common password cracking tools?

Answer: John the Ripper, Hashcat, Hydra

Category: Vulnerabilities & Threats

Key Takeaway: John the Ripper and Hashcat are mainly used to crack password hashes, while Hydra is commonly used to test login credentials against network services.

Question 3

Question: What is the Kerberos authentication process?

Answer: Kerberos is an authentication protocol that uses encrypted tickets instead of repeatedly sending a user's password across the network. First, the user logs in and receives a Ticket Granting Ticket (TGT). The TGT is then used to request a service ticket, which is presented to the server to prove the user's identity and gain access.

Category: Access Control & Permissions

Key Takeaway: Kerberos follows the basic process Login → TGT → Service Ticket → Access and helps protect passwords by using tickets for authentication.

Date: 8-14-2026

Question 1

Question: What is a mount point?

Answer: A mount point is a directory in Linux where a storage device or file system is attached so its files can be accessed. For example, a USB drive might be mounted to a directory such as /mnt/usb.

Category: Linux

Key Takeaway: A mount point is the location in the Linux directory structure where a file system or storage device becomes accessible.

Question 2

Question: What is an NBNS/LLMNR poisoning attack?

Answer: An NBNS/LLMNR poisoning attack happens when an attacker responds to local network name-resolution requests before the legitimate system does. The victim may then connect to the attacker's machine and send authentication information, such as an NTLM challenge-response, which the attacker may try to capture or relay.

Category: Vulnerabilities & Threats

Key Takeaway: NBNS/LLMNR poisoning tricks devices into trusting a malicious system during local name resolution, which can expose or relay Windows authentication credentials.

Question 3

Question: What mechanism is commonly used to push configuration changes to Windows servers?

Answer: Group Policy is commonly used to centrally deploy and enforce configuration changes across Windows computers and servers in an Active Directory environment. Administrators create or update Group Policy Objects (GPOs), which are then applied to selected users, computers, organizational units, or domains.

Category: Windows

Key Takeaway: Group Policy and GPOs let administrators centrally push and enforce Windows configuration changes across multiple systems.

Date: 8-17-2026

Question 1

Question: What is a Virtual Machine

Answer: A Virtual Machine (VM) is a software-based computer that runs its own operating system and applications while sharing the physical hardware of a host system. VMs are created and managed by a hypervisor.

Category: Cloud & Virtualization

Key Takeaway: A VM acts like an independent computer, but it runs virtually on shared physical hardware.

Question 2

Question: What is east-west VM communication?

Answer: East-west VM communication refers to network traffic moving between virtual machines or systems within the same internal environment, such as within a data center or cloud network.

Category: Cloud & Virtualization

Key Takeaway: East-west traffic stays inside the environment, while north-south traffic generally moves into or out of the environment.

Question 3

Question: What is IaaS? (Infrastructure as a Service)

Answer: Infrastructure as a Service (IaaS) is a cloud computing model where a provider supplies virtualized resources such as servers, storage, and networking. The customer typically manages the operating systems, applications, and configurations running on that infrastructure.

Category: Cloud & Virtualization

Key Takeaway: IaaS provides the underlying computing infrastructure while giving the customer control over the operating systems and applications.

Date: 8-18-2026

Question 1

Question: What hypervisor has a smaller attack surface, Type 1 or Type 2? Why?

Answer: A Type 1 hypervisor generally has a smaller attack surface because it runs directly on the physical hardware instead of on top of a host operating system. A Type 2 hypervisor depends on the security of both the hypervisor software and the underlying host OS, creating more components that could potentially be attacked.

Category: Cloud & Virtualization

Key Takeaway: Type 1 hypervisors usually have a smaller attack surface because they eliminate the extra host operating system layer.

Question 2

Question: Why is Ring -1 important to protect?

Answer: Ring -1 refers to the hypervisor layer, which sits below the operating system and controls virtual machines and access to the underlying hardware. If an attacker compromises the hypervisor, they could potentially affect multiple VMs at once, bypass protections inside individual operating systems, and gain very powerful control over the virtual environment.

Category: Cloud & Virtualization

Key Takeaway: Ring -1 is critical to protect because compromising the hypervisor can give an attacker control over multiple virtual machines and the systems running on them.

Question 3

Question: Out of the Threat Mitigation Framework layers, which layer is most important for preventing VM Escape?

Answer: Hypervisor hardening is the most important layer for preventing VM Escape. VM Escape occurs when an attacker breaks out of a guest virtual machine and gains access to the hypervisor or host system, so securing and patching the hypervisor directly addresses that risk.

Category: Cloud & Virtualization

Key Takeaway: Because VM Escape targets the boundary between the VM and the hypervisor, hypervisor hardening is the primary defense.

Date: 8-19-2026

Question 1

Question: What is the difference between Type 1 and Type 2 Hypervisors?

Answer: A Type 1 hypervisor runs directly on the physical hardware, while a Type 2 hypervisor runs on top of a normal operating system. Type 1 hypervisors are generally used in enterprise environments because they offer better performance, security, and resource management.

Category: Cloud & Virtualization

Key Takeaway: Type 1 runs directly on hardware, while Type 2 depends on a host operating system.

Question 2

Question: What is the difference between a virtual machine and a container?

Answer: A virtual machine includes its own full operating system and runs through a hypervisor, while a container shares the host system's operating system kernel and mainly packages the application and its dependencies. Containers are generally more lightweight and faster to start than VMs.

Category: Cloud & Virtualization

Key Takeaway: VMs virtualize entire operating systems, while containers virtualize applications while sharing the host OS kernel.

Question 3

Question: What is a jump host?

Answer: A jump host, or jump server, is a secured system that administrators connect to first before accessing sensitive servers or internal systems. It acts as a controlled gateway and helps limit, monitor, and log administrative access.

Category: Access Control & Permissions

Key Takeaway: A jump host provides a secure, centralized entry point for accessing sensitive systems instead of allowing direct administrative connections.

Date: 8-20-2026

Question 1

Question: What are 4 ways we can have shared storage?

Answer: Four common shared storage methods are SAN (Storage Area Network), NAS (Network Attached Storage), iSCSI, and Fibre Channel. These allow multiple servers or hosts to access centralized storage instead of relying only on local disks.

Category: Cloud & Virtualization

Key Takeaway: Shared storage lets multiple hosts access the same centralized data, which is important for clustering, migration, and high availability.

Question 2

Question: What is the difference between Fault Tolerance (FT) and High Availability (HA)?

Answer: Fault Tolerance keeps a duplicate system running at the same time so service can continue with little or no interruption if the primary system fails. High Availability detects a failure and restarts or moves the workload to another available host, which may cause a short period of downtime.

Category: Cloud & Virtualization

Key Takeaway: FT aims for near-zero downtime, while HA focuses on quick recovery after a failure.

Question 3

Question: What is the difference between a bastion host and a jump host?

Answer: A jump host is a controlled system administrators use to access other internal systems. A bastion host is a specially hardened system designed to be exposed to or

positioned near an untrusted network and act as a secure access point. In practice, a bastion host can also function as a jump host.

Category: Access Control & Permissions

Key Takeaway: A jump host focuses on controlling administrative access, while a bastion host emphasizes hardened security at the boundary of a trusted network.

Date: 8-21-2026

Question 1

Question: What is Infrastructure as Code (IaC)?

Answer: Infrastructure as Code is the practice of managing and configuring infrastructure through code or configuration files instead of making changes manually. It allows environments to be deployed consistently, repeatedly, and automatically.

Category: Cloud & Virtualization

Key Takeaway: IaC makes infrastructure easier to automate, standardize, and recreate consistently.

Question 2

Question: How do you isolate a machine in a virtual environment?

Answer: A virtual machine can be isolated by placing it on a separate virtual network, VLAN, or subnet and restricting communication with firewall rules, access controls, or virtual switch settings. The goal is to limit what other systems the VM can communicate with.

Category: Cloud & Virtualization

Key Takeaway: VM isolation uses network segmentation and access controls to prevent unnecessary communication between systems.

Question 3

Question: What is lateral movement?

Answer: Lateral movement is when an attacker who has already compromised one system moves through the network to access additional systems, accounts, or resources. Attackers often use stolen credentials, remote access tools, or trusted network connections to move between systems.

Category: Vulnerabilities & Threats

Key Takeaway: Lateral movement allows an attacker to expand from one compromised system to other parts of the network.

Date: 8-24-2026

Question 1

Question: What is the difference between a MAC address and an IP address?

Answer: A MAC address is a hardware address used to identify a network interface on the local network, while an IP address is a logical address used to identify a device and route traffic between networks. MAC addresses operate mainly at Layer 2, while IP addresses operate at Layer 3.

Category: Networking

Key Takeaway: MAC addresses identify devices locally, while IP addresses are used to route traffic across networks.

Question 2

Question: What is ARP poisoning, and why is ARP vulnerable to it?

Answer: ARP poisoning is an attack where a malicious device sends false ARP information so other systems associate the attacker's MAC address with another device's IP address, such as the default gateway. ARP is vulnerable because it does not authenticate ARP replies and devices can accept unsolicited updates.

Category: Vulnerabilities & Threats

Key Takeaway: ARP poisoning works because ARP trusts mapping updates without verifying that they are legitimate.

Question 3

Question: What information does a computer need from DHCP to communicate properly on a network?

Answer: DHCP commonly provides the computer with an IP address, subnet mask, default gateway, and DNS server. These settings allow the device to identify itself, determine which devices are local, reach other networks, and resolve domain names.

Category: Networking

Key Takeaway: The four main DHCP settings to remember are IP address, subnet mask, default gateway, and DNS server.

Date: 8-25-2026

Question 1

Question: What is the difference between RBAC and ABAC?

Answer: RBAC (Role-Based Access Control) gives permissions based on a user's assigned role, such as administrator, analyst, or employee. ABAC (Attribute-Based Access Control) makes access decisions using attributes such as department, location, device type, clearance level, or time of day.

Category: Access Control & Permissions

Key Takeaway: RBAC uses roles, while ABAC uses attributes and conditions to make more detailed access decisions.

Question 2

Question: What is the difference between SSO, MFA, and a one-time password?

Answer: SSO (Single Sign-On) allows a user to authenticate once and access multiple approved applications. MFA (Multi-Factor Authentication) requires two or more different

authentication factors, such as a password and a phone notification. A one-time password (OTP) is a temporary code that can only be used once and is often used as one factor in MFA.

Category: Access Control & Permissions

Key Takeaway: SSO reduces the number of logins, MFA increases authentication security, and an OTP is a temporary credential that can be used as part of MFA.

Question 3

Question: What are the most important MFA enforcement and recovery strategies?

Answer: MFA should be enforced for primary logins, privileged or administrative actions, and remote access. Common enforcement methods include conditional access policies, IAM-integrated requirements, and application-level enforcement. Organizations should also maintain secure recovery options such as backup codes, controlled recovery workflows, and carefully protected break-glass accounts.

Category: Access Control & Permissions

Key Takeaway: Strong MFA requires both consistent enforcement across important systems and secure recovery methods that do not create an easy way to bypass MFA.

Date: 8-26-2026

Question 1

Question: What is the difference between encryption, hashing, and HMAC?

Answer: Encryption converts plaintext into ciphertext using a key and is designed to be reversible with the correct key. Hashing converts data into a fixed-length digest and is designed to be one-way. HMAC combines a cryptographic hash function with a shared secret key to verify both the integrity and authenticity of a message.

Category: General Cybersecurity

Key Takeaway: Encryption protects confidentiality, hashing helps verify integrity, and HMAC verifies both integrity and authenticity.

Question 2

Question: What is the difference between symmetric, asymmetric, and hybrid encryption?

Answer: Symmetric encryption uses the same secret key to encrypt and decrypt data and is generally very fast. Asymmetric encryption uses a public and private key pair and is useful for secure key exchange and digital signatures. Hybrid encryption combines both methods by using asymmetric encryption to securely exchange a symmetric key, then using symmetric encryption for the actual data.

Category: General Cybersecurity

Key Takeaway: Symmetric encryption is fast, asymmetric encryption helps establish trust and exchange keys securely, and hybrid encryption combines the strengths of both.

Question 3

Question: What is PKI, and how do digital certificates and signatures establish trust?

Answer: PKI (Public Key Infrastructure) is the system of certificates, public/private keys, certificate authorities, and policies used to establish trusted digital identities. A digital certificate connects a public key to a verified identity, while a digital signature uses a private key to prove who signed the data and confirm that it has not been altered.

Category: General Cybersecurity

Key Takeaway: PKI creates a trusted system for managing digital identities, while certificates verify who owns a public key and digital signatures provide authenticity, integrity, and non-repudiation.

Date: 8-31-2026

Question 1

Question: What is the difference between a traditional firewall and a next-generation firewall (NGFW)?

Answer: A traditional firewall mainly makes decisions based on information such as IP addresses, ports, protocols, and connection state. A next-generation firewall (NGFW)

adds deeper inspection features such as application awareness, intrusion prevention, threat intelligence, identity-based policies, and SSL/TLS inspection.

Category: General Cybersecurity

Key Takeaway: Traditional firewalls focus mostly on where traffic is coming from, where it is going, and what port it uses, while NGFWs can also inspect what the traffic actually is and whether it behaves maliciously.

Question 2

Question: How does an NGFW detect malicious traffic that is hidden inside normal HTTPS traffic on port 443?

Answer: An NGFW can use SSL/TLS inspection to decrypt HTTPS traffic, inspect its contents and behavior, and then re-encrypt it before forwarding it. It can also use application identification, intrusion prevention, and behavioral analysis to detect traffic that looks like normal HTTPS but is actually malware or command-and-control communication.

Category: Vulnerabilities & Threats

Key Takeaway: An NGFW does not automatically trust traffic just because it uses HTTPS or port 443; it can decrypt and inspect the traffic to determine what it is actually doing.

Question 3

Question: Why are firewall rules and configuration just as important as the firewall's technical capabilities?

Answer: Advanced firewall features only provide protection when the rules and policies are configured to use them correctly. Poorly written or overly permissive rules can allow unauthorized traffic, lateral movement, or data exfiltration even when the firewall itself has advanced security capabilities.

Category: System Hardening

Key Takeaway: A powerful firewall is only as effective as its configuration; bad rules can make advanced security features ineffective.

Date: 9-1-2026

Question 1

Question: Why does the order of firewall rules matter?

Answer: Firewall rules are usually processed from top to bottom, and the first matching rule is applied. Because of this, a broad rule placed above a more specific rule can cause the specific rule to never be reached.

Category: System Hardening

Key Takeaway: Firewall rule order matters because the first matching rule usually wins, so more specific rules should be placed carefully to avoid unintended access or blocking.

Question 2

Question: What is the difference between default deny and default allow?

Answer: Default deny blocks traffic unless it is specifically permitted by a rule. Default allow permits traffic unless it is specifically blocked. Default deny is generally more secure because only approved traffic is allowed through.

Category: System Hardening

Key Takeaway: Default deny = block everything unless approved; default allow = allow everything unless blocked.

Question 3

Question: What is a shadow rule, and why can it be dangerous?

Answer: A shadow rule is a firewall rule that never gets used because an earlier rule already matches the same traffic. This can be dangerous because administrators may think the shadowed rule is providing protection when it is actually being ignored.

Category: System Hardening

Key Takeaway: A shadow rule is hidden behind a broader earlier rule, which can create unexpected access, security gaps, and difficult troubleshooting.

Date: 9-2-2026

Question 1

Question: Why are virtual firewall appliances useful in cloud and virtualized environments?

Answer: Virtual firewall appliances provide firewall capabilities without requiring dedicated physical hardware. They can be deployed inside cloud or virtual environments to inspect traffic between virtual machines, networks, and applications, making them easier to scale and place where traffic actually flows.

Category: Cloud & Virtualization

Key Takeaway: Virtual firewalls bring security controls directly into virtual and cloud environments where physical firewalls may not see all traffic.

Question 2

Question: Why should firewall rule updates and network segmentation be used together?

Answer: Firewall rule updates help block known threats, while network segmentation limits how far an attacker can move if something gets through. Using both provides layered protection because one control helps cover the weaknesses of the other.

Category: System Hardening

Key Takeaway: Updated rules block known threats, while segmentation limits the damage from unknown or successful attacks.

Question 3

Question: What are common signs of firewall evasion?

Answer: Common signs include traffic using unusual or non-standard ports, repeated blocked connection attempts, unexpected protocol behavior, suspicious encrypted

traffic, port scanning, and applications trying to disguise themselves as normal allowed traffic.

Category: Vulnerabilities & Threats

Key Takeaway: Firewall evasion often appears as unusual ports, repeated denied traffic, protocol anomalies, or malicious traffic trying to blend in with legitimate connections.

Date: 9-3-2026

Question 1

Question: Why is centralized management important in a multi-LAN environment?

Answer: Centralized management gives administrators one place to monitor multiple LANs, apply consistent security policies, and make configuration changes. It also helps prevent policy drift, where different networks slowly end up with different or outdated security settings.

Category: Networking

Key Takeaway: Centralized management improves visibility, consistency, and control across all connected LANs.

Question 2

Question: How does network segmentation reduce the risk of lateral movement?

Answer: Network segmentation divides a network into smaller zones or segments and controls the traffic allowed between them. If an attacker compromises one system, segmentation can prevent or limit them from freely moving into other parts of the network.

Category: Networking

Key Takeaway: Segmentation limits an attacker's ability to move from one compromised area to another, reducing the potential blast radius of an attack.

Question 3

Question: Why should traffic be monitored at every LAN interconnection point?

Answer: LAN interconnection points are places where traffic moves from one network segment to another, making them important locations for detecting suspicious activity. Monitoring these points can reveal unauthorized access, unusual traffic patterns, malware communication, and attempts at lateral movement.

Category: Networking

Key Takeaway: Monitoring every LAN interconnection helps prevent visibility gaps and makes it easier to detect threats moving between networks.

Date: _____

Question 1

Question:

Answer:

Category:

Key Takeaway:

Question 2

Question:

Answer:

Category:

Key Takeaway:

Question 3

Question:

Answer:

Category:

Key Takeaway:

Quick Reference

Important Ports:

- 20/21 - FTP: File transfers. Port 21 handles commands, and 20 is traditionally used for data. Not secure because standard FTP does not encrypt traffic.
- 22 - SSH: Secure remote login and administration. Also used by SFTP.
- 23 - Telnet: Remote login, but unencrypted and considered insecure. SSH is the safer replacement.
- 25 - SMTP: Used to send email between mail servers.
- 53 - DNS: Translates domain names like google.com into IP addresses.
- 67/68 - DHCP: Automatically assigns IP addresses and network settings to devices.
- 80 - HTTP: Standard unencrypted web traffic.
- 110 - POP3: Used to retrieve email from a mail server. Standard POP3 is unencrypted.
- 143 - IMAP: Used to access and manage email stored on a mail server.
- 443 - HTTPS: Secure, encrypted web traffic. One of the most important ports to know.
- 636 - LDAPS: Secure version of LDAP (Lightweight Directory Access Protocol). It is used to securely access and manage directory services such as user accounts, groups, and authentication information.
-

Important Commands:

- ls - Lists files and directories in the current location.
- ls -la - Shows all files, including hidden files, along with permissions and ownership.
- cd - Changes directories.
- pwd - Shows your current directory.
- cat - Displays the contents of a file.
- grep - Searches for specific text inside files or command output.
- find - Searches for files and directories.
- chmod - Changes file and directory permissions.
- chown - Changes the owner or group of a file or directory.
- sudo - Runs a command with elevated administrator privileges.
- sudo -l - Shows which commands the current user is allowed to run with sudo.
- id - Shows a user's UID, GID, and group memberships.

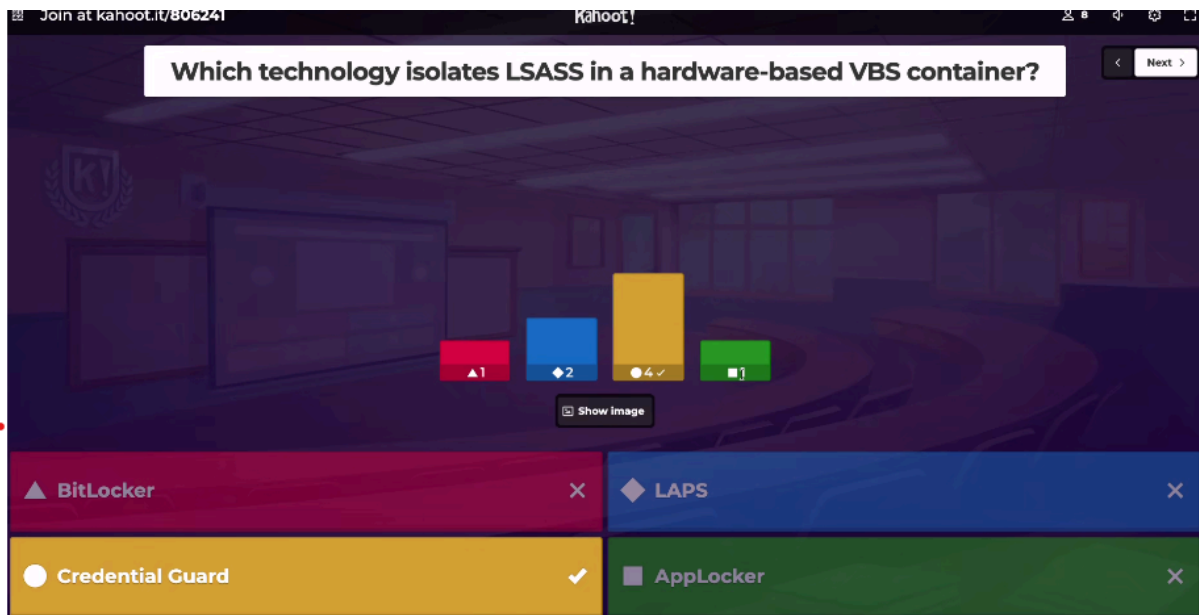
- whoami - Shows the currently logged-in username.
- cat /etc/passwd - Lists local user accounts and account information.
- passwd - Changes a user's password.
- sudo passwd -l username - Locks a user account.
- ps aux - Shows running processes.
- systemctl status service - Checks the status of a service.
- systemctl list-units --type=service --state=running - Lists currently running services.
- sudo systemctl disable --now service - Stops a service and prevents it from starting automatically.
- ss -tulpn - Shows listening ports and the services using them.
- ipconfig - Shows network interfaces and IP addresses.
- ping - Tests connectivity to another system.
- nslookup - Looks up DNS information for a domain.
- curl - Sends requests to websites or web services and displays the response.
- ssh user@host - Connects securely to a remote Linux system.
- apt update - Refreshes the list of available packages and updates.
- apt upgrade - Installs available package updates.
- apt list --upgradable - Shows packages that have updates available.
- history - Displays previously entered commands.
- man command - Opens the manual page for a command.
- clear - Clears the terminal screen.
-

Important Frameworks / Standards:

- NIST Cybersecurity Framework (CSF) - Helps organizations manage cybersecurity risk using the functions Govern, Identify, Protect, Detect, Respond, and Recover.
- NIST SP 800-53 - A large catalog of security and privacy controls commonly used by government agencies and organizations with strong security requirements.
- NIST SP 800-171 - Focuses on protecting Controlled Unclassified Information (CUI) in non-federal systems and organizations.
- ISO/IEC 27001 - International standard for building and maintaining an Information Security Management System (ISMS).
- CIS Controls - A prioritized set of practical cybersecurity safeguards organizations can implement to reduce common attacks.
- CIS Benchmarks - Detailed hardening recommendations for systems such as Ubuntu, Debian, RHEL, Windows, and cloud platforms.
- PCI DSS - Security standard for organizations that store, process, or transmit payment card information.
- SOC 2 - Auditing framework focused on how service organizations protect customer data, especially around security, availability, confidentiality, processing integrity, and privacy.
- HIPAA Security Rule - Requires safeguards to protect electronic protected health information in healthcare environments.

- SOX - U.S. law focused on financial reporting and internal controls, including IT controls that affect financial data.
 - Zero Trust Architecture - Security approach based on the idea of never automatically trusting users or devices, even if they are already inside the network.
-

Missed Kahoots



Which feature ties encryption to a user's certificate rather than the machine?

< Next >

▲1 ✓ ◆3 ●2 ■2

▲ EFS ✓	◆ TLS ✕
● SMB Encryption ✕	■ VBS ✕

Which registry value must be set to 0 to disable WDigest?

< Next >

▲1 ◆3 ●2 ✓ ■2

▲ RunAsPPL ✕	◆ RestrictAnonymous ✕
● UseLogonCredential ✓	■ NoDriveTypeAutoRun ✕

Join at kahoot.it/806241 Kahoot!

Which registry value, set to 1, enables LSA Protection for LSASS?

2 1 4 1 ✓

▲ UseLogonCredential	✕	◆ RestrictAnonymousSAM	✕
● EnableSMB1Protocol	✕	■ RunAsPPL	✓

Join at kahoot.it/806241 Kahoot!

What should be enforced, not just enabled, to prevent SMB signing downgrades?

2 4 ✓ 1 1

▲ Enable SMB encryption	✕	◆ Require SMB signing	✓
● Disable SMBv1	✕	■ Enable TLS 1.3	✕

Which default administrative shares should be removed if not needed for IT ops?

▲ 3 ◆ 3 ● 2 ✓ ■ 0

▲ SYSVOL and NETLOGON	×	◆ Public and Users	×
● C\$ and ADMIN\$	✓	■ Backup and Logs	×

Which command checks for a pending reboot on Red Hat systems?

▲ 2 ◆ 2 ● 1 ■ 3 ✓

▲ apt list --upgradable	×	◆ systemctl status	×
● ss -tulnp	×	■ needs-restarting -r	✓

Join at kahoot.it/806241 Kahoot!

Which command removes a service from boot but still allows manual starting?

▲ 0 ◆ 3 ● 5 ✓ ■ 0

▲ systemctl mask	×	◆ systemctl stop	×
● systemctl disable	✓	■ systemctl status	×

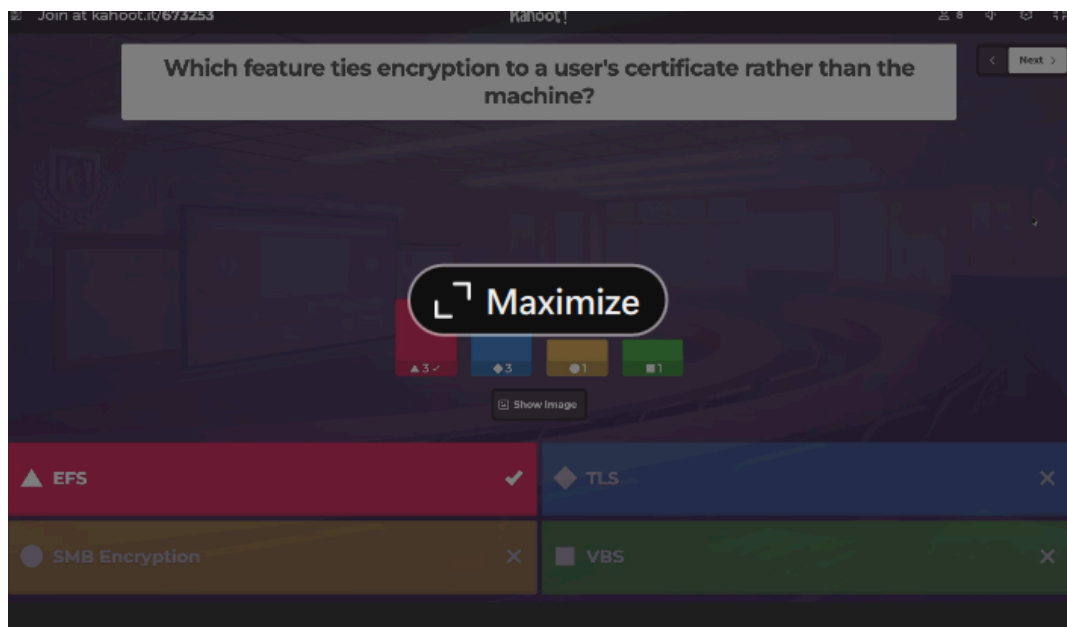
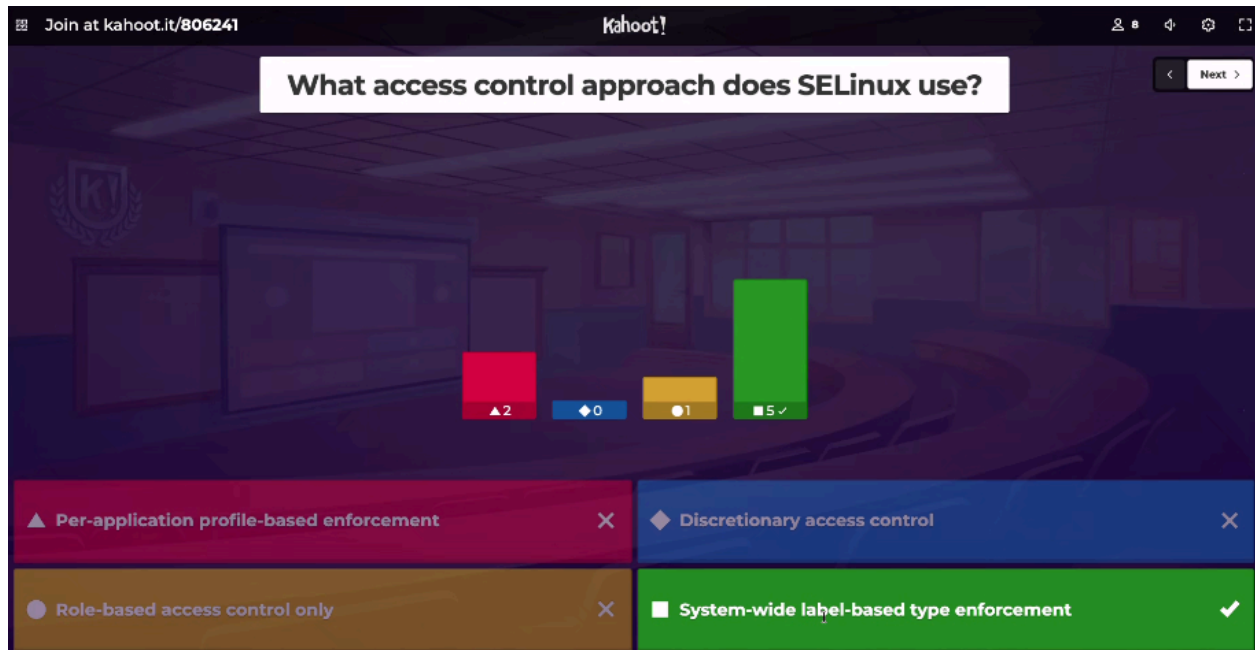
Join at kahoot.it/806241 Kahoot!

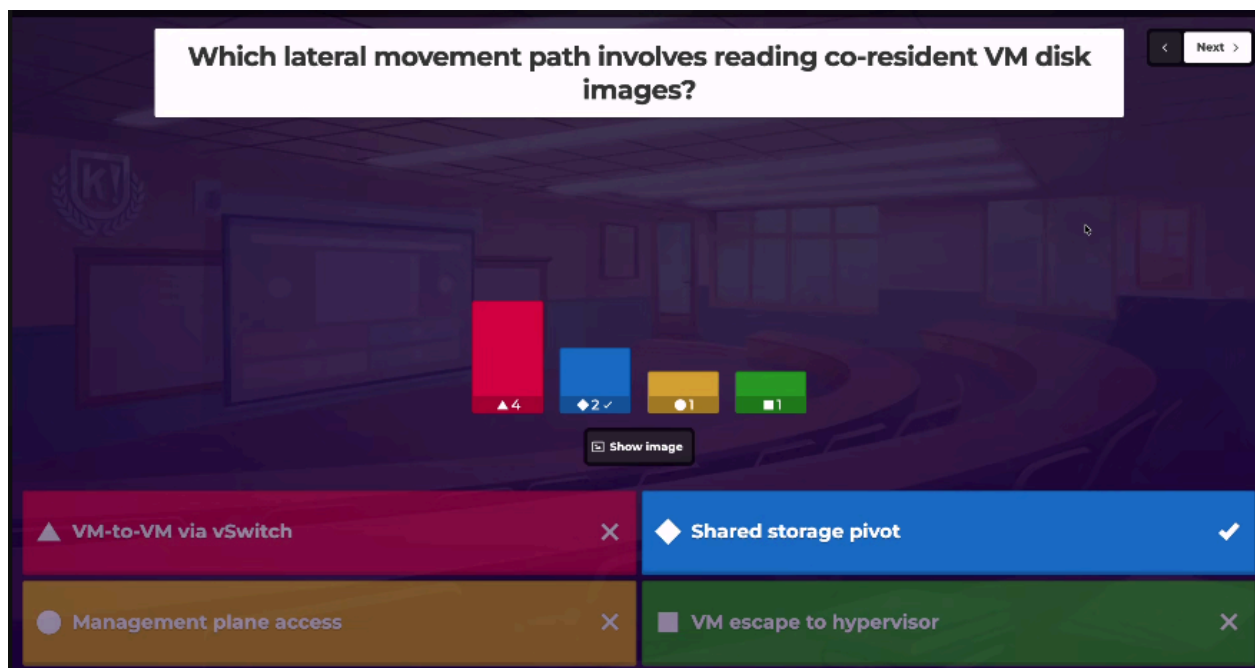
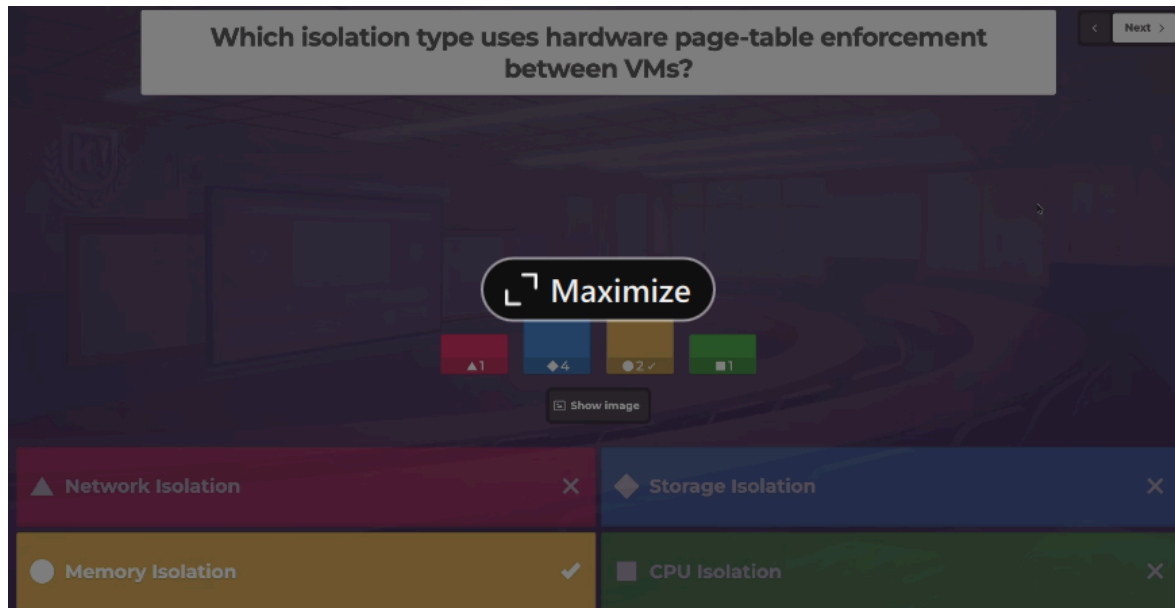
Which command applies systemctl config files without requiring a reboot?

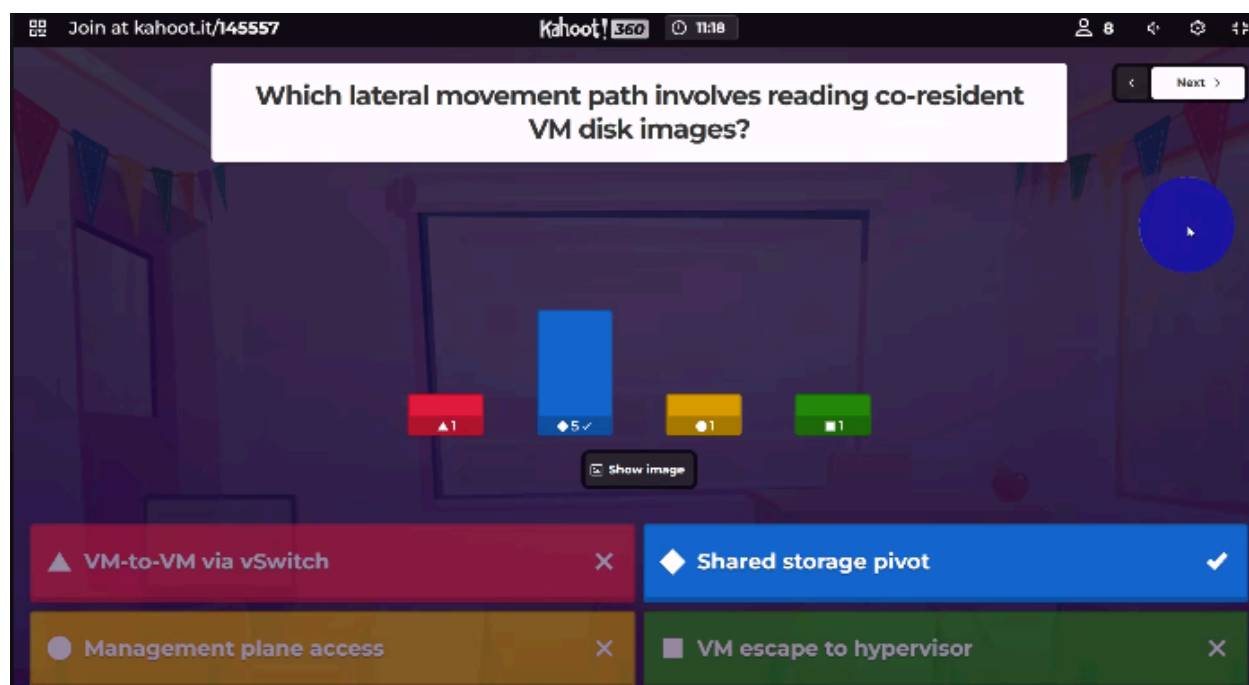
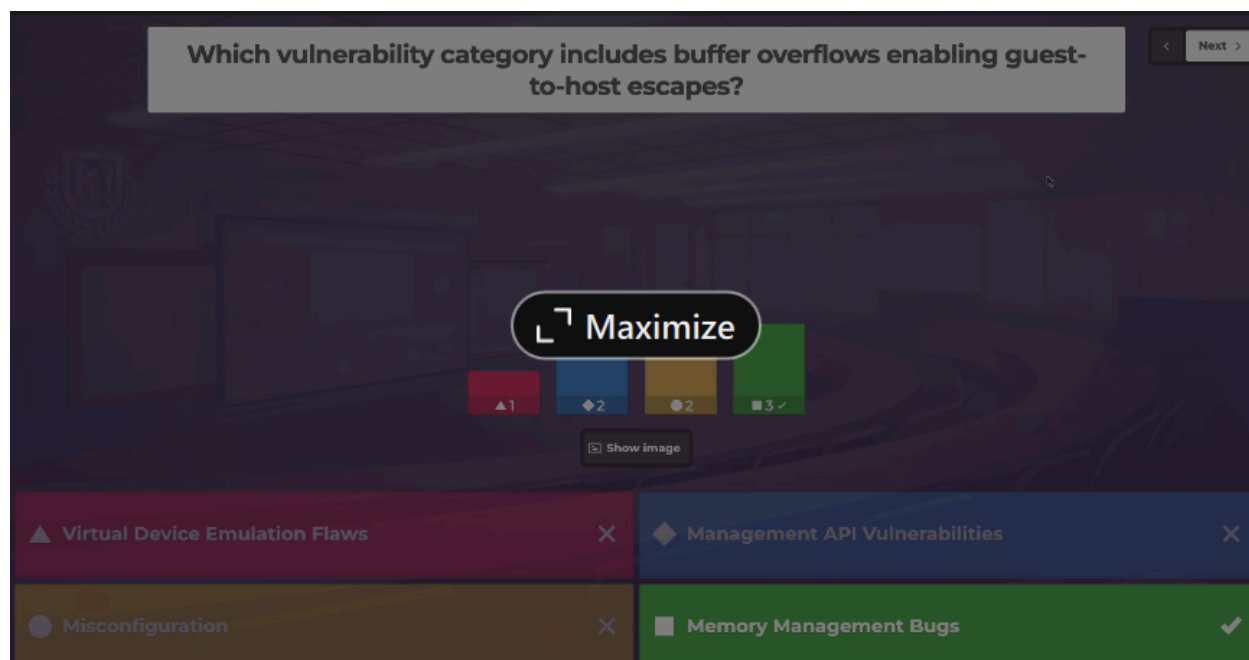
▲ 1 ◆ 3 ✓ ● 1 ■ 3

▲ systemctl -w	×	◆ systemctl --system	✓
● systemctl -a		■ reload	×

Stop video (Ctrl + Shift + V)







Join at kahoot.it/839331 Kahoot! 360 12:37 8 < Next >

Which hardware feature lets the CPU do guest-to-host address translation?

▲ 2 ◆ 0 ● 2 ■ 3 ✓

Show image

▲ Transparent Page Sharing	×	◆ Memory Ballooning	×
● NUMA	×	■ Extended Page Tables (EPT/NPT)	✓

Join at kahoot.it/839331 Kahoot! 360 11:59 8 < Next >

Which memory technique inflates a guest driver to force paging under pressure?

▲ 5 ✓ ◆ 1 ● 0 ■ 2

Show image

▲ Memory Ballooning	✓	◆ Transparent Page Sharing	×
● Extended Page Tables	×	■ Memory Overcommit	×

Which technology keeps a synchronized shadow VM for zero-downtime failover?

▲1 ◆0 ●2 ■5✓

Show Image

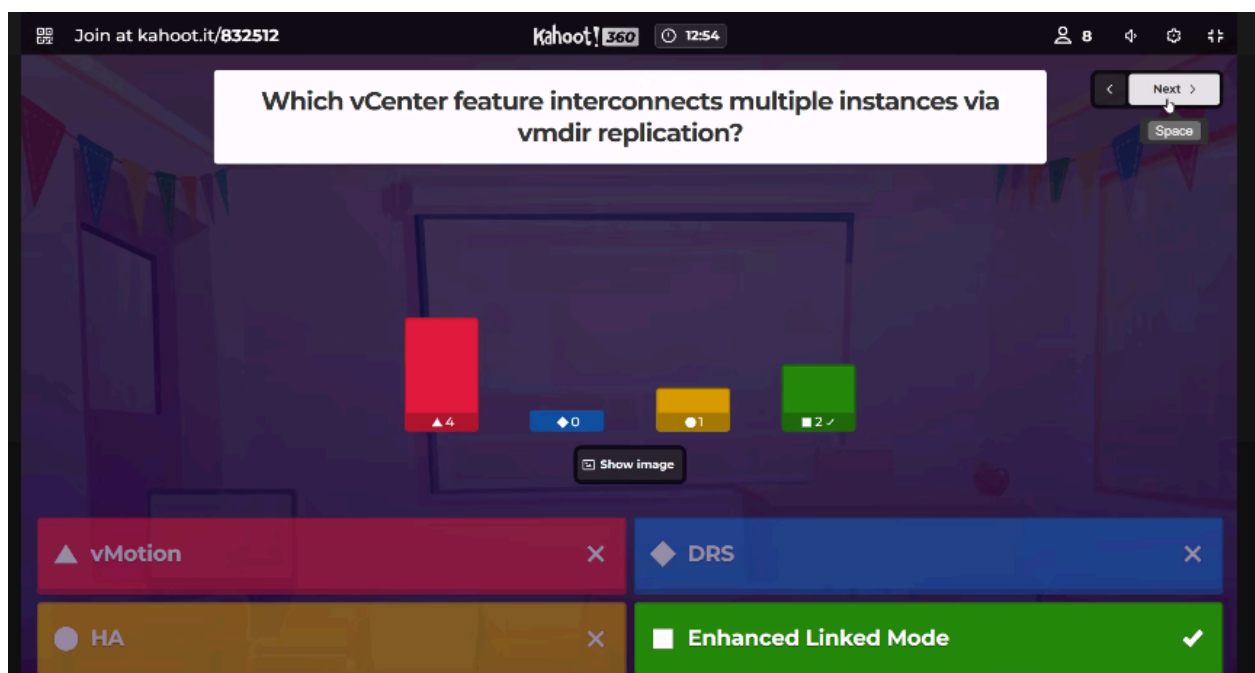
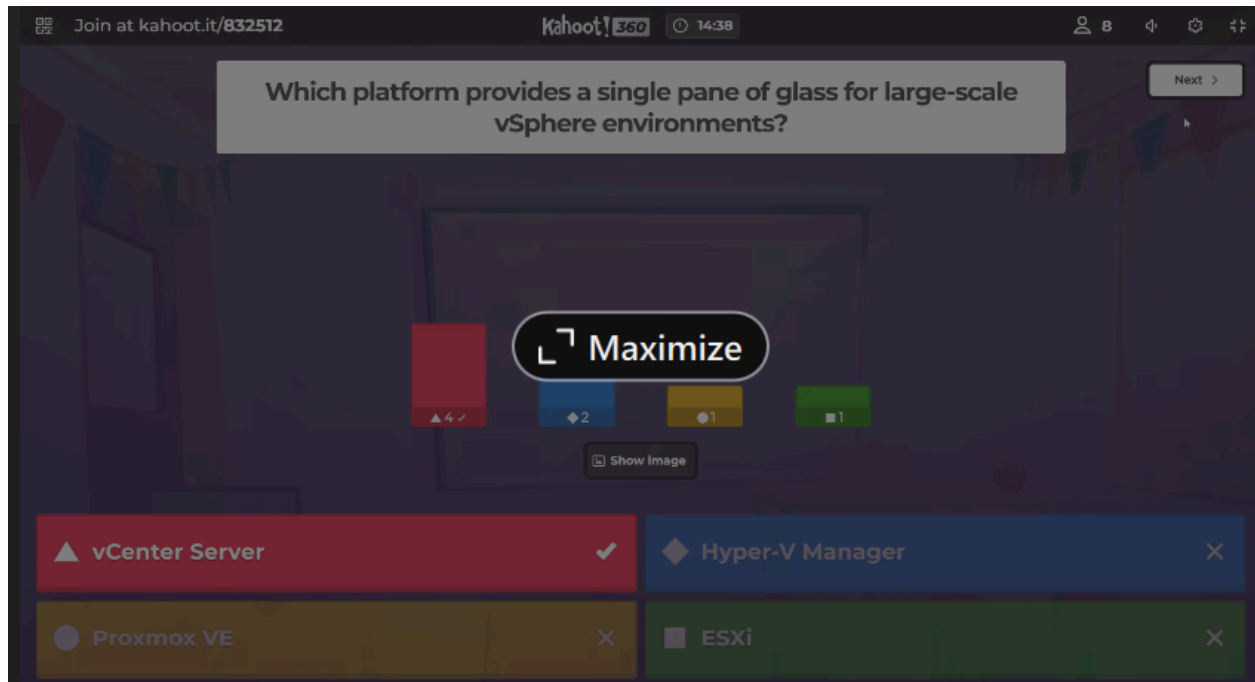
▲ High Availability (HA)	×	◆ vMotion	×
● Hyper-V Replica	×	■ Fault Tolerance (FT)	✓

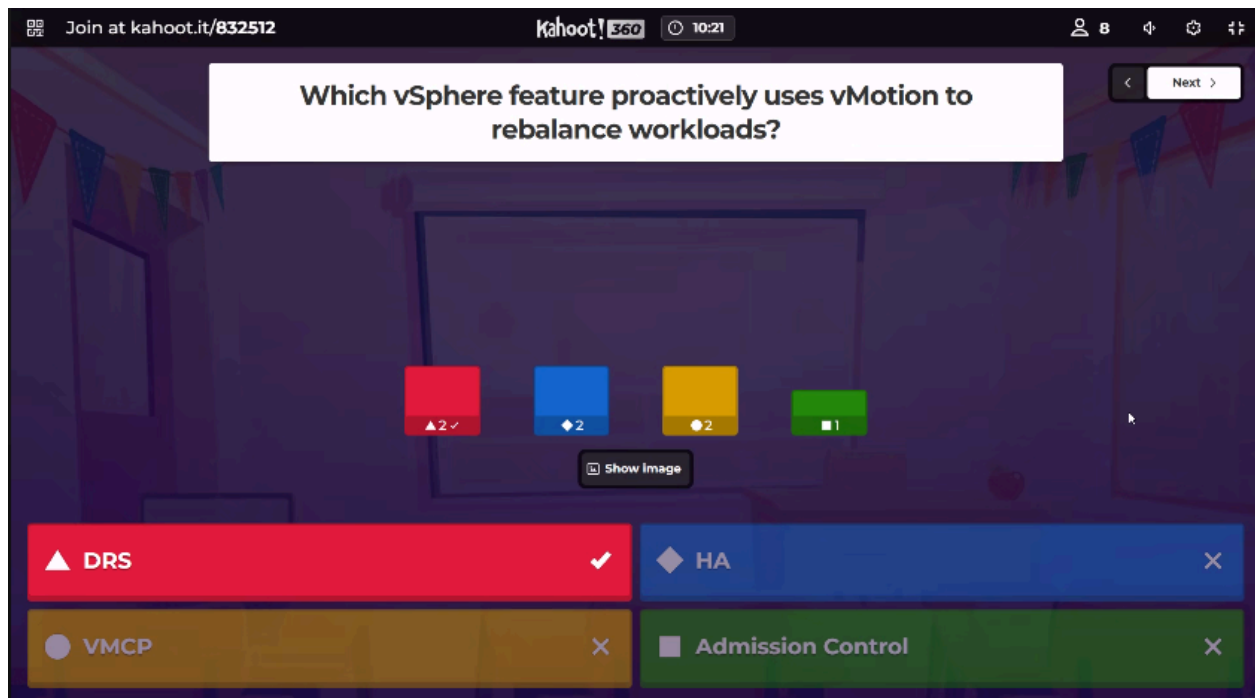
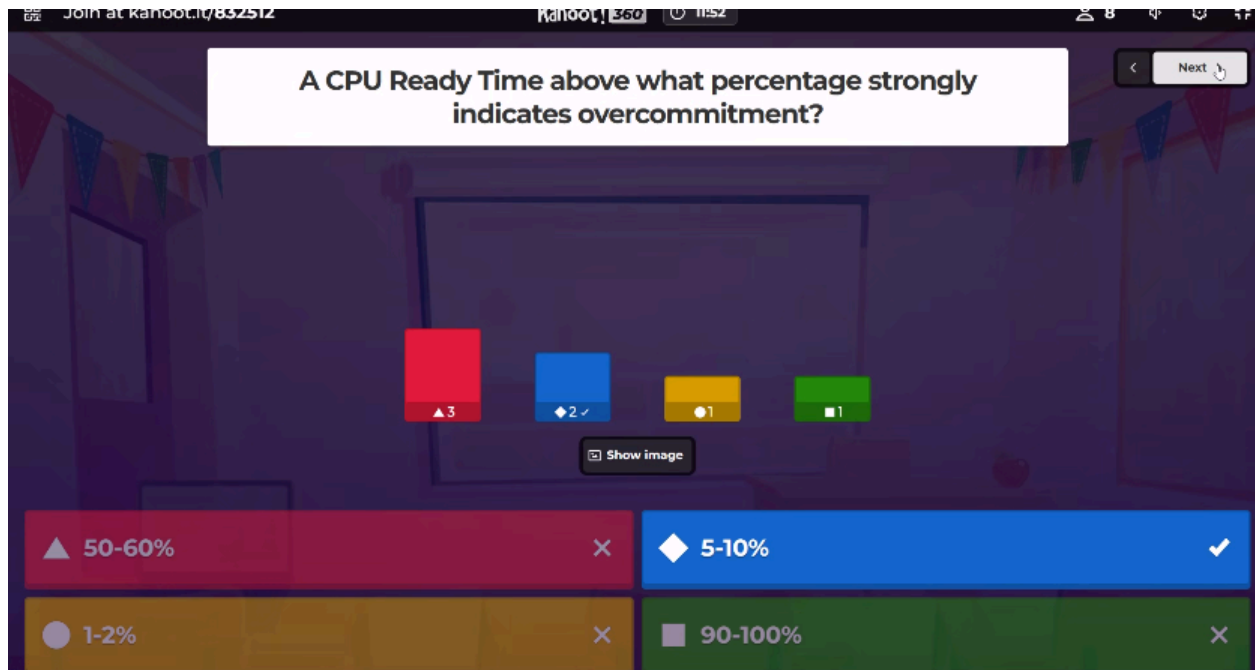
What must be isolated from cardholder data networks per PCI DSS guidance?

▲4 ◆3✓ ●0 ■1

Show Image

▲ All guest VM traffic	×	◆ Hypervisor management interfaces	✓
● Backup storage only	×	■ DNS servers	×





Join at kahoot.it/832512 Kahoot! 360 4:37 8

What does vSphere HA require so any surviving host can restart VMs?

Next >

▲ 4 ◆ 2 ● 2 ✓ ■ 0

Show Image

▲ Enhanced Linked Mode	×	◆ CPU overcommitment	×
● Shared storage	✓	■ Per-VM licensing	×

Join at kahoot.it/183258 Kahoot! 360 13:56 7

Which pre-configured Linux appliance is VMware's standard vCenter deployment?

Next >

▲ 2 ◆ 2 ● 1 ✓ ■ 2

Show Image

▲ ESXi ISO	×	◆ SCVMM	×
● vCenter Server Appliance (VCSA)	✓	■ vSphere Client	×

Join at kahoot.it/184810 Kahoot! 360 14:12 7

A hypervisor or management plane compromise is only visible at which layer?

Next >

▲ 2 ◆ 3 ✓ ● 2 ■ 0

Show image

▲ VM-Level Monitoring	✕	◆ Host-Level Monitoring	✓
● Perimeter monitoring	✕	■ DNS monitoring	✕

Join at kahoot.it/245020 Kahoot! 360 14:30 7

Which protocol type shares its full routing table with neighbors, trusting them?

Next >

▲ 1 ✓ ◆ 3 ● 2 ■ 1

Show image

▲ Distance-Vector	✓	◆ Link-State	✕
● Path-Vector	✕	■ Static	✕

Join at kahoot.it/245020 Kahoot! 360 13:14 7

What is the hard hop limit that makes RIP unsuitable for large networks?

▲ 1 ◆ 1 ● 4 ✓ ■ 1

Show image

▲ 30 hops	✕	◆ 10 hops	✕
● 15 hops	✓	■ 255 hops	✕

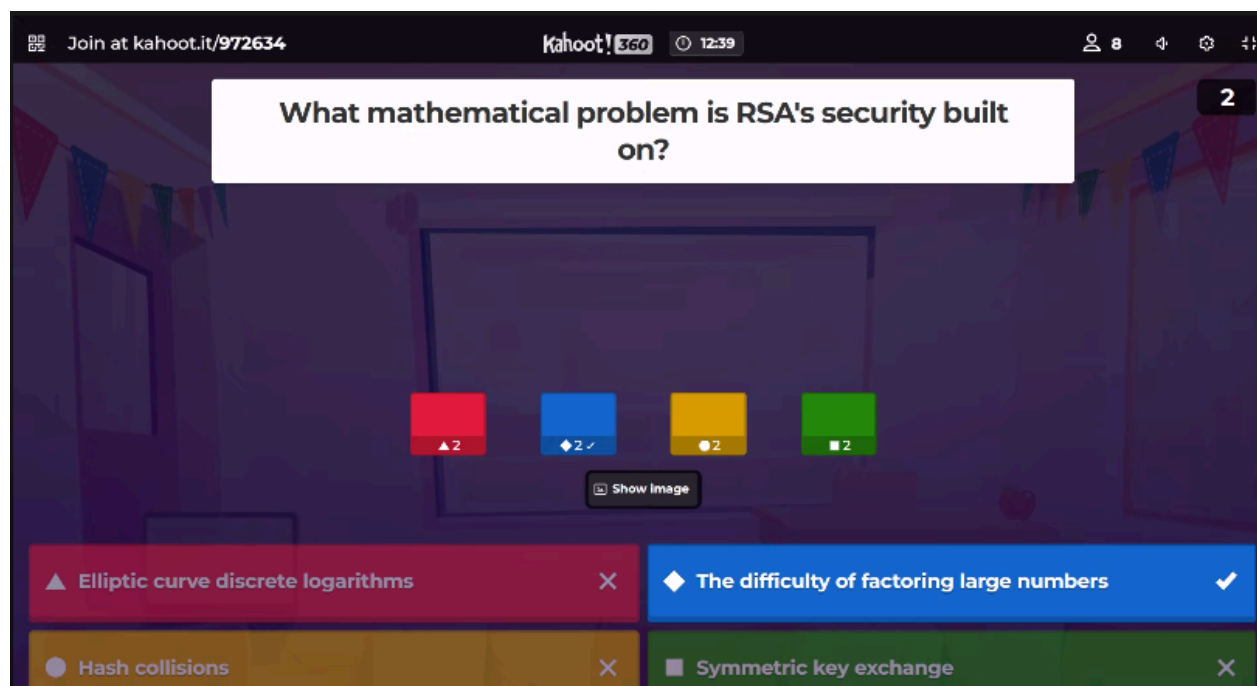
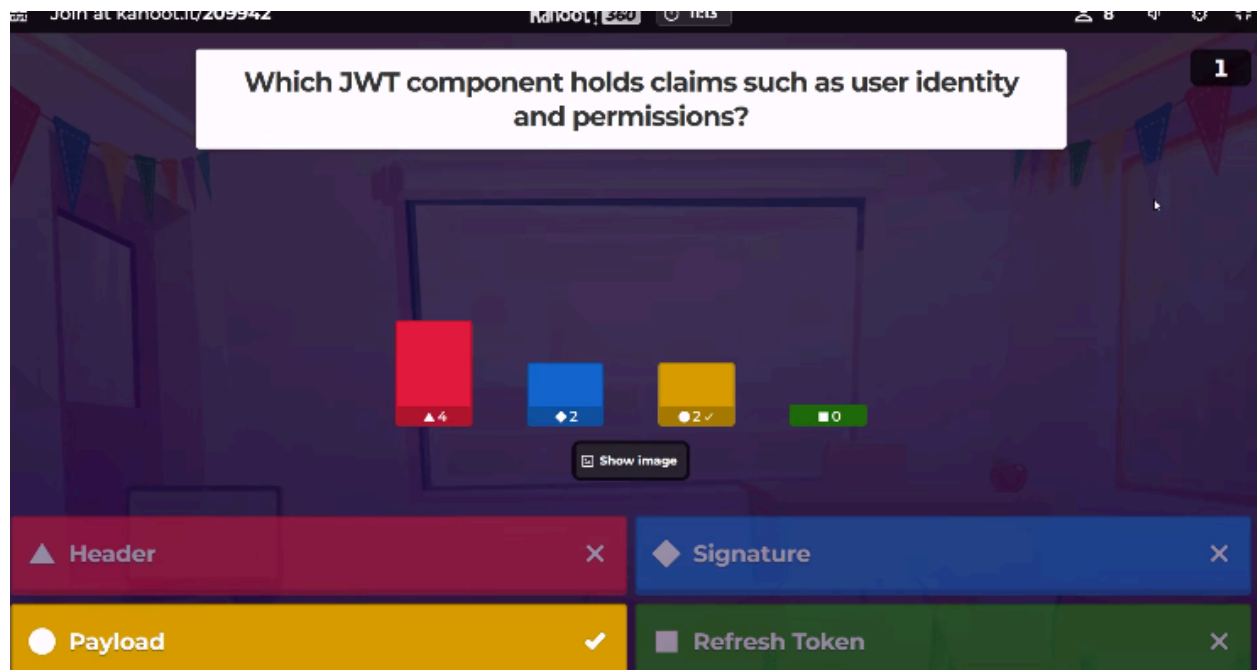
Join at kahoot.it/245020 Kahoot! 360 9:41 7

Which IP header field prevents routing loops by decrementing at each hop?

▲ 0 ◆ 2 ● 2 ■ 3 ✓

Show image

▲ Source address	✕	◆ Destination address	✕
● Fragment offset	✕	■ TTL (Time to Live)	✓



Join at kahoot.it/972634 Kahoot! 360 9:15 8

Which key is used to verify a digital signature? 1

▲ 3 ✓ ◆ 1 ● 1 ■ 3

Show image

▲ The signer's public key ✓	◆ The signer's private key ✕
● A shared symmetric key ✕	■ The recipient's private key ✕

Join at kahoot.it/11771 Kahoot! Plus 8

Which deployment model mirrors traffic and cannot block anything? < Next >

▲ 4 ◆ 0 ● 2 ✓ ■ 2

Show image

▲ Inline ✕	◆ Active-active ✕
● Out-of-Band ✓	■ Active-passive ✕

